

OOPS ASSIGNMENT

Question1.

Code :

```
#include <iostream>

using namespace std;

class Rectangle{
public:
    int length, breadth;

    Rectangle(){};

    Rectangle(int a, int b){
        length = a; breadth = b;
    }

    Rectangle(int x){
        length = x;
        breadth = x;
    }

    void area(){
        int area = length*breadth;
        cout << area;
    };
};

int main(){

    Rectangle A2(79,7); A2.area();

    cout << " " ; Rectangle A1(67); A1.area();

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
553 4489
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question2.

```
#include <iostream>

using namespace std;

class Rectangle {
    int length, breadth;

public:
    Rectangle() {
        length = 0; breadth = 0;
        cout << "Default constructor called\n";
    }

    Rectangle(int l, int b) {
        length = l;
        breadth = b;
        cout << "Parameterized constructor called\n";
    }

    ~Rectangle() {
        cout << "Destructor called\n";
    }

    int area() {
        return length * breadth;
    }

};

int main() {
    Rectangle arr[3] = { Rectangle(), Rectangle(5, 4),

    Rectangle(7, 3)

};

    for (int i = 0; i < 3; i++) {
```

```
cout << "Area of object " << i+1 << ": " << arr[i].area() << endl;
}

return 0;

}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Default constructor called
Parameterized constructor called
Parameterized constructor called
Area of object 1: 0
Area of object 2: 20
Area of object 3: 21
Destructor called
Destructor called
Destructor called
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```

Question3.

i. True

ii. True

iii. True

iv. True

Code:

```
#include <iostream>

using namespace std;

class Rectangle{

public:

    int length, breadth;

    Rectangle(){

        cout<<"Constructor is called"<<endl;

    };

    ~Rectangle(){

        cout<< "Destructor is called"<< endl;

    };

};

int main(){

    Rectangle();
```

```
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ tempCodeRunnerFile.CPP -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Constructor is called
Destructor is called
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Code:

```
#include <iostream>

using namespace std;

class Demo{
public:
    Demo(){
        cout<<"Constructor is called"<<endl;
    };
    ~Demo(){
        cout<< "Destructor is called"<< endl;
    };
};

int main(){
    Demo obj;
    return 0;
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPS> cd "c:\Users\vaibh\OneDrive\Desktop\OOPS\" ; if ($?) { g++ tempCodeRunnerFile.CPP -o tempCodeRunnerFile } ; if ($?) { .\tempCodeRunnerFile }
Constructor is called
Destructor is called
PS C:\Users\vaibh\OneDrive\Desktop\OOPS>
```

Question4.

Code:

```
#include <iostream>

using namespace std;

class Test {
public:
    int x;
    Test(int val) { x = val;
```

```
}  
  
void show() {  
    cout << "Value: " << x << endl;  
}  
  
};  
  
int main() {  
    // Integer  
    int *p = new int;  
    *p = 10;  
    cout << "Integer: " << *p << endl; delete p;  
  
    // Float  
    float *f = new float;  
    *f = 5.5;  
    cout << "Float: " << *f << endl; delete f;  
  
    // Integer array  
    int *arr = new int[3]; for(int i = 0; i < 3; i++) {  
        arr[i] = i + 1;  
        cout << arr[i] << " ";  
    }  
    cout << endl; delete[] arr;  
  
    // Object  
    Test *obj = new Test(100); obj->show();  
    delete obj;  
  
    // Array of objects  
    Test *objArr = new Test[2]{Test(1), Test(2)}; for(int i = 0; i < 2; i++) {  
        objArr[i].show();  
    }  
    delete[] objArr; return 0;  
}
```

Output:

```
PS C:\Users\vaibh\OneDrive\Desktop\OOPs> cd "c:\Users\vaibh\OneDrive\Desktop\OOPs\" ; if ($?) { g++ ASSIGNMENT.CPP -o ASSIGNMENT } ; if ($?) { .\ASSIGNMENT }
Integer: 10
Float: 5.5
1 2 3
Value: 100
Value: 1
Value: 2
PS C:\Users\vaibh\OneDrive\Desktop\OOPs>
```